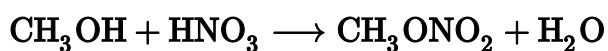


Nitrate ester

In organic chemistry, a **nitrate ester** is an organic functional group with the formula $R-ONO_2$, where R stands for any organyl group. They are the esters of nitric acid and alcohols. A well-known example is nitroglycerin, which is not a *nitro* compound, despite its name.

Synthesis and reactions

Nitrate esters are typically prepared by condensation of nitric acid and the alcohol:^{[1][2]} For example, the simplest nitrate ester, methyl nitrate, is formed by reaction of methanol and nitric acid in the presence of sulfuric acid.^[3]

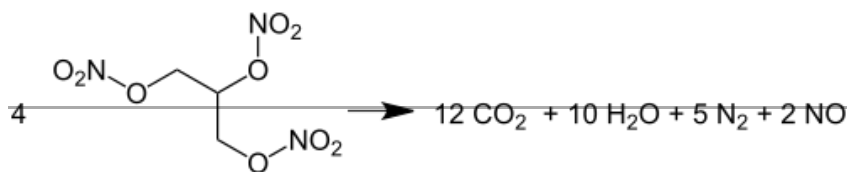


Formation of a nitrate ester is called a nitrooxylation (less commonly, nitroxylation).

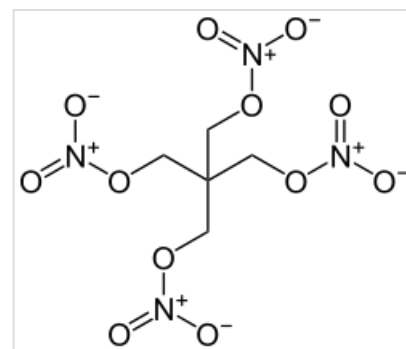
Most commonly, "mixed acid" (nitric and sulfuric acids) are used, but in the 1980s production of the nitrocellulose with magnesium nitrate as a dehydrating agent was started in the US.^[4] In laboratory, phosphoric acid and phosphorus pentoxide or acetic acid and its anhydride may be used for the same purpose, or the nitroxylation can be conducted in anhydrous conditions (such as dichloromethane or chloroform).^[5]

Explosive properties

The thermal decomposition of nitrate esters mainly yields the gases molecular nitrogen (N_2) and carbon dioxide. The considerable chemical energy of the detonation is due to the high strength of the bond in molecular nitrogen. This stoichiometry is illustrated by the equation for the detonation of nitroglycerin.



Illustrative of the highly sensitive nature of some organic nitrates is $\text{Si}(\text{CH}_2\text{ONO}_2)_4$.^{[6][7]} A single crystal of this compound detonates even upon contact with a teflon spatula and in fact made full characterization impossible. Another contributor to its exothermic decomposition (inferred from much safer in silico experimentation) is the ability of silicon in its crystal phase to coordinate to two oxygen nitrito groups in addition to regular coordination to the four carbon atoms. This additional coordination would make formation of silicon dioxide (one of the decomposition products) more facile.



Pentaerythritol tetranitrate is a commercially important explosive that contains four nitrate ester groups.

Medical use

The nitrate esters isosorbide dinitrate (Isordil) and isosorbide mononitrate (Imdur, Ismo, Monoket, Mononitron) are converted in the body to nitric oxide, a potent natural vasodilator. In medicine, these esters are used as a medicine for angina pectoris (ischemic heart disease).

Related compounds

Acetyl nitrate is a nitrate anhydride, being derived from the condensation of nitric and acetic acids.

Atmospheric Occurrence

Atmospheric Abundance

Alkyl nitrates (RONO₂) compounds are widespread in the boundary layer, accounting for a large portion of total reactive nitrogen oxides in both urban and rural environments and across various hydrocarbon mixtures.^[8] In addition, alkyl nitrates are typically complex, multifunctional molecules that are still challenging to detect using chromatographic methods at the concentrations found under ambient conditions. Alkyl nitrates have been found to contribute to the NO_y compound budget. In total, 43 (C₆–C₁₃) alkyl mononitrates, 24 (C₃–C₆) alkyl dinitrates, and 19 (C₂–C₆) hydroxy alkyl nitrates were identified. In the urban air samples, the summed concentrations of 15 (C₆–C₁₀) alkyl mononitrates ranged from 2.9 to 11.0 pptv. The total levels of 21 (C₃–C₆) alkyl dinitrates were between 2.3 and 10.5 pptv, and the sum of 7 (C₂–C₄) hydroxy alkyl nitrates ranged from 7.3 to 28 pptv. These findings indicate that alkyl dinitrates, hydroxy alkyl nitrates, and alkyl mononitrates make an important contribution to the NO_y compound budget. Moreover, no significant differences in either the concentrations or patterns of organic nitrates were observed between the urban air of cities.^[9] Organic nitrates, however, occur at comparatively low concentrations. Short-chain alkyl nitrates (C₄–C₆) exhibit mixing ratios of about 0.2–2.5 pptv, with a pronounced minimum in the tropics and much lower levels in the Southern Hemisphere. For the longer and more functionalized species, the summed mixing ratio of 36 long-chain alkyl mononitrates (C₇–C₁₃) lies between 0.02 and 0.43 pptv, that of 23 alkyl dinitrates (C₃–C₆) ranges from 0.005 to 1.08 pptv, and that of 7 hydroxy alkyl nitrates (C₂–C₄) ranges from 0.005 to 1.07 pptv.^[10]

Alkyl Nitrates Abundance in Variant Geographic Regions

Concentrations (pptv) / Region	Urban area	Marine air
Alkyl mononitrates	2.9 – 11.0	0.02 – 0.43
Alkyl dinitrates	2.3 – 10.5	0.005 – 1.08
Hydroxy alkyl nitrates	7.3 – 28	0.005 – 1.07

Formation Mechanisms

There are two primary mechanisms for producing RONO₂ in the atmosphere:

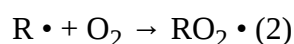
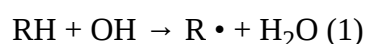
(1) oxidation of hydrocarbons initiated by hydroxyl radicals (OH) in the presence of NO_x during the daytime.

(2) oxidation of alkenes initiated by nitrate radicals (NO₃) at night.

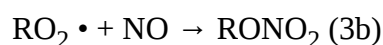
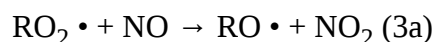
Methyl nitrate (CH₃ONO₂), ethyl nitrate (C₂H₅ONO₂), and possibly propyl nitrate (C₃H₇ONO₂) are released directly from the oceans, and concentrations of these compounds reaching up to several tens of ppt have been observed in remote marine environments.^[11] Apart from these marine emissions, direct emissions are generally not regarded as a major source. RONO₂ exhibit a wide range of vapor pressures. It occurs both in gaseous and aerosol particles.^[12]

OH-Initiated RONO₂ Formation

When OH reacts with a saturated hydrocarbon (1), it first abstracts a hydrogen atom, and almost immediately afterward, oxygen (O₂) adds to the resulting radical to form an RO₂ radical (2).



Two distinct product pathways (3a and 3b) arise from the reaction between RO₂ and NO when NO_x is present.

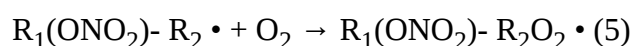
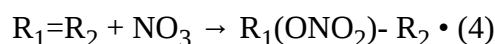


The dominant pathway 3a, produces NO₂ and an alkoxy radical and thereby continues the NO_x and HO_x catalytic cycles. In contrast, the less frequent pathway 3b, terminates the chain by generating a stable monofunctional organic nitrate. Reaction 3b is a termolecular process and represents only a minor pathway (typically <5%) compared with the major, bimolecular pathway, reaction 3a.

NO₃-Initiated RONO₂ Formation

During daylight hours, the OH-initiated pathway described previously is the primary source of RONO₂. At night, in contrast, RONO₂ formation occurs through reactions of NO₃ with alkenes and phenols.^[13] These nighttime reactions have high nitrate formation amounts. Although the NO₃ pathway constitutes only a small portion of overall organic oxidation relative to OH, this pathway can still account for up to about 50% of the regional RONO₂ burden.^[14]

The process begins with NO₃ adding to a carbon–carbon double bond (4). The alkyl radical formed in this step then rapidly reacts with O₂, analogous to reaction 2, yielding a peroxy radical (5).



Once formed, this peroxy radical can react with HO₂, RO₂, or NO₃, and these reactions yield stable aldehyde- or alcohol-nitrate products. In contrast, pathways involving NO (3a and 3b) are negligible in the atmosphere because NO₃ cannot accumulate where NO is present; NO and NO₃ rapidly react together, producing two NO₂ molecules. Nitrooxy-peroxynitrates (R₁(ONO₂)–R₂O₂NO₂) can arise from the reaction of nitrooxy-peroxy radicals with NO₂ and have been detected as transient species in laboratory experiments.^[15] However, their strong thermal instability and the lack of ambient detections indicate that they mainly influence how laboratory results are interpreted rather than playing a direct role in atmospheric chemistry. NO₃ is also theoretically capable of oxidizing alkanes through hydrogen atom abstraction, but this reaction proceeds so slowly that it is not considered important under atmospheric conditions.

Atmospheric Removal

For gas phase, a RONO₂ molecule has several possible fates: it may be transported by atmospheric winds, take part in further chemical reactions, deposit onto the surface, or partition into the aerosol phase. How quickly any of these processes occur depends on the molecule's specific structure, which controls both its reactivity and its tendency to deposit or move into particles.

Nitrates that contain multiple functional groups, either formed that way in the first oxidation step or made multifunctional after successive reactions, are expected to be particularly reactive. Molecules that still possess a carbon–carbon double bond or have abstractable hydrogen atoms also tend to be processed more rapidly. As one example, first-generation isoprene nitrates react quickly with both O₃ and OH, and all eight isomers could survive daytime oxidation but no more than a few hours. With continued oxidation, the rate of resulting higher-generation products would differ from those of the parent nitrate.

The detailed products formed when RONO₂ is oxidized by OH, O₃, and NO₃ are not well constrained. The overall chemistry can be thought as two main pathways:

1. Oxidation preserves the nitrate group and produces a stable multifunctional nitrate, R₁ONO₂. Here the original carbon skeleton R has been altered so that the new backbone R₁ carries additional functional groups (e.g., carbonyls, alcohols, or extra nitrate groups).
2. Oxidation removes the nitrate functionality, releasing NO₂ and effectively stripping the nitrate group from the molecule.

RONO₂ compounds absorb near-UV radiation ($\lambda < 340$ nm) and break down through photodissociation, yielding RO• radicals and NO₂ with quantum efficiencies close to unity. Experimental studies have determined photolysis rates for numerous C₁–C₅ alkyl and cycloalkyl nitrates^[16], as well as for several difunctional organic nitrates.^[17]

For smaller members (C₁–C₄), photolysis rates rival oxidation by OH. Under typical summer surface conditions, photolytic lifetimes can vary from 3 to 10 days. On the other hand, most RONO₂ species photolysis much slower than reactions with the peroxy radical. Because of this, photolysis is not significant to atmospheric loss.^[18]

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